Direct Energy Conversion in the S.M.A.R.T. Architecture

Performance, Modal Diversity, and System Integration

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Abstract

Direct energy conversion (DEC) in aneutronic fusion promises to bypass the thermal Carnot limits of conventional cycles by directly harvesting kinetic energy of charged fusion products and high-energy photons as electricity. This paper presents a comprehensive engineering review of DEC for spherical tokamak reactors fueled by advanced aneutronic fuels (D–^{3}He and p–^{11}B). Multiple complementary DEC pathways are examined, including electrostatic multi-stage ion collectors, magnetohydrodynamic (MHD) generation from charged plasma exhaust, high-temperature thermionic emission using refractory hexaboride emitters, and radiative photovoltaic conversion via advanced GaAs/perovskite cells. Integration of these direct converters with high-temperature Brayton cycles (He or sCO_2) for residual thermal photon energy is analyzed to achieve maximal overall efficiency. The theoretical performance bounds and practical design considerations are discussed, drawing on historic DEC experiments (Moir's Venetian blind converters on TMX, the UW "Apollo"-L3 D–^{3}He reactor study) and modern developments. A hybrid topping architecture is outlined in which front-end DEC captures the bulk of charged-particle energy while a Brayton bottoming cycle converts the remaining heat, yielding total efficiencies well beyond those of Rankine systems. Applications ranging from ultra-compact space power units to grid-scale generators for high-demand computing are considered. The results underscore DEC's strategic potential to enable high-efficiency, neutron-free fusion reactors that are compact, high-output, and well-suited for AI-era power needs.

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Introduction

Fusion reactions that predominantly release energy in charged particles (aneutronic fuels) open the door to **direct energy conversion (DEC)**, wherein kinetic energy is converted straight to electricity without intermediate thermalization. This is a compelling alternative to conventional Rankine steam cycles, which are fundamentally limited by Carnot efficiency and massive heat-transfer hardware. In aneutronic fusion, up to ~80–100% of the reaction energy appears as kinetic energy of charged ions or as prompt electromagnetic radiation, rather than neutrons[1][2]. This makes it feasible to "skip" the thermal stage and directly convert energy at far higher theoretical efficiency. Prior studies have projected that multi-stage electrostatic converters could recover on the order of 60–70% of fusion product energy[3][4], whereas even advanced steam cycles (e.g. helium-cooled SiC turbines at ~1000 °C) reach only ~40–50%[5]. In principle, with ideal direct conversion of all charged outputs, a D–^{3}He or p–^{11}B reactor could approach *double* the net efficiency of a comparable thermal-only plant.

Early experimental validation of DEC was achieved in the 1970s–80s. Notably, Moir and Barr's **"Venetian blind"** direct converter concept – a multi-grid ion energy separator – demonstrated 50–65% energy capture in beam tests[6]. A single-stage plasma direct converter tested on the Tandem Mirror Experiment (TMX) at Livermore directly converted escaping plasma energy with about **48% efficiency**[6][7], a remarkable proof-of-concept. Marshall Rosenbluth cautioned that maintaining space-charge neutrality over the short deceleration distance would be challenging in larger devices[8], highlighting a key engineering issue. Nonetheless, these results confirmed the viability of harvesting electricity straight from fusion exhaust. In parallel, design studies like the University of Wisconsin's **Apollo-L3** tokamak reactor embraced DEC. Apollo-L3 was a 1000 MWe D–^{3}He design that combined direct conversion of synchrotron radiation (via rectifying antennas) with a thermal cycle[9]. It achieved a **net efficiency of 47%**, significantly higher than a comparable D–T reactor's ~39%[5][10]. The direct conversion of >60% of the D–^{3}He plasma's synchrotron power at ~80% efficiency contributed to this gain[9]. These comparisons illustrate DEC's promise: for example, using DEC to *topping-cycle* a fusion plant could push net electric efficiency beyond Fifty or even 60%, vs. ~35–40% for a pure steam cycle[5][11].

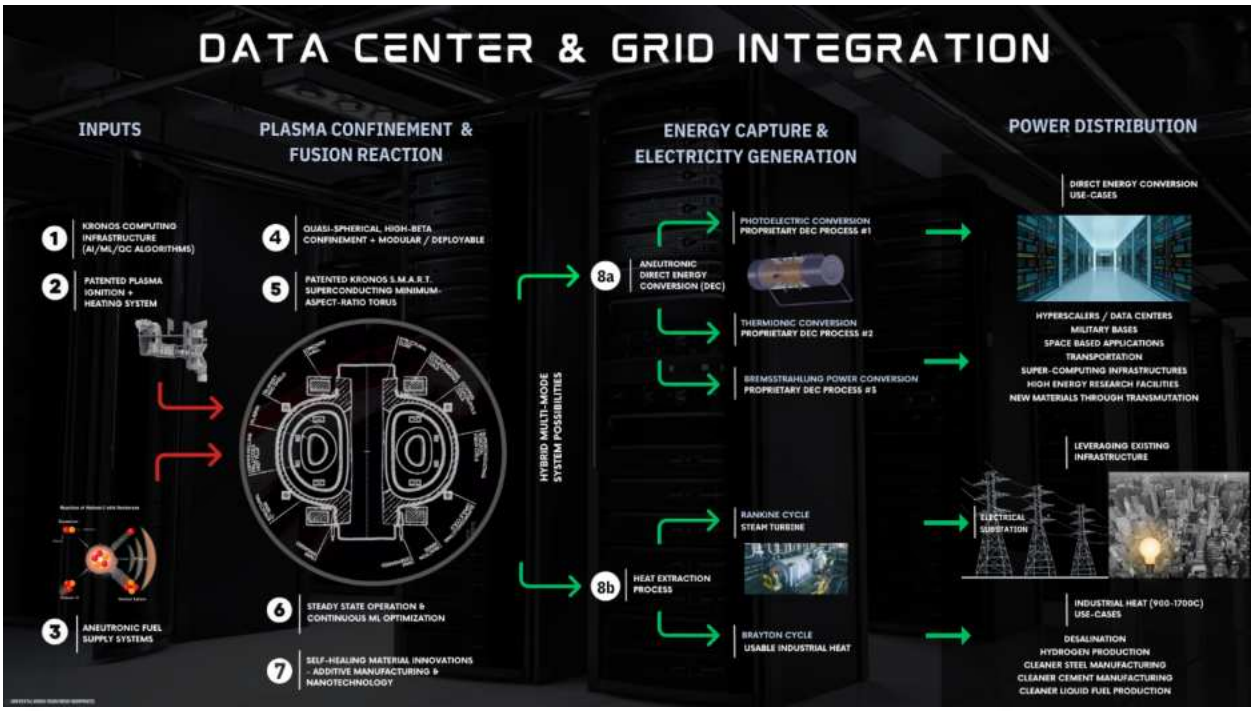

Figure 1: High-level schematic of energy capture in the S.M.A.R.T. architecture. Charged particles are routed to electrostatic or MHD converters, photons are harvested via radiovoltaics and Brayton thermal loops, and residual heat is recovered through a Rankine cycle for maximal energy utilization (≥70 % potential net efficiency).

This paper focuses on practical engineering pathways to implement DEC in spherical tokamaks using advanced fuels. We first summarize the energy outputs of **D–^{3}He and p–^{11}B** reactions and their compatibility with direct conversion. We then delve into four DEC modalities: (A) electrostatic direct collectors, (B) magnetohydrodynamic (MHD) generators, (C) thermionic converters, and (D) photoelectric (radiovoltaic) systems. For each, we discuss operating principles, design considerations (e.g. space-charge limits, material constraints), and recent developments. Section 5 examines how these modalities can be integrated in a hybrid architecture with a high-temperature Brayton cycle to capture any remaining thermal/photon energy. We provide an energy balance analysis and an illustrative efficiency table comparing a DEC-only system, a hybrid DEC+Brayton system, and a traditional Rankine cycle. Finally, Section 6 outlines prospective applications – from compact space power units and mobile generators to grid-scale fusion plants for data centers – highlighting how multi-modal energy outputs can be managed in future energy systems. The conclusion underlines the strategic value of DEC in realizing ultra-efficient, neutron-free fusion reactors that align with emerging needs for compact, on-demand power.

Energy Output of Aneutronic Fuels

Aneutronic fusion fuels release the bulk of their energy in charged reaction products rather than neutrons. Two leading candidates are the D–^{3}He (deuterium–helium-3) reaction and the p–^{11}B (proton–boron-11) reaction. **D–^{3}He fusion** proceeds as:

$$\mathrm{D} + {}^3\mathrm{He} \rightarrow {}^4\mathrm{He}\ (3.6\ \mathrm{MeV}) + \mathrm{p}\ (14.7\ \mathrm{MeV}),$$

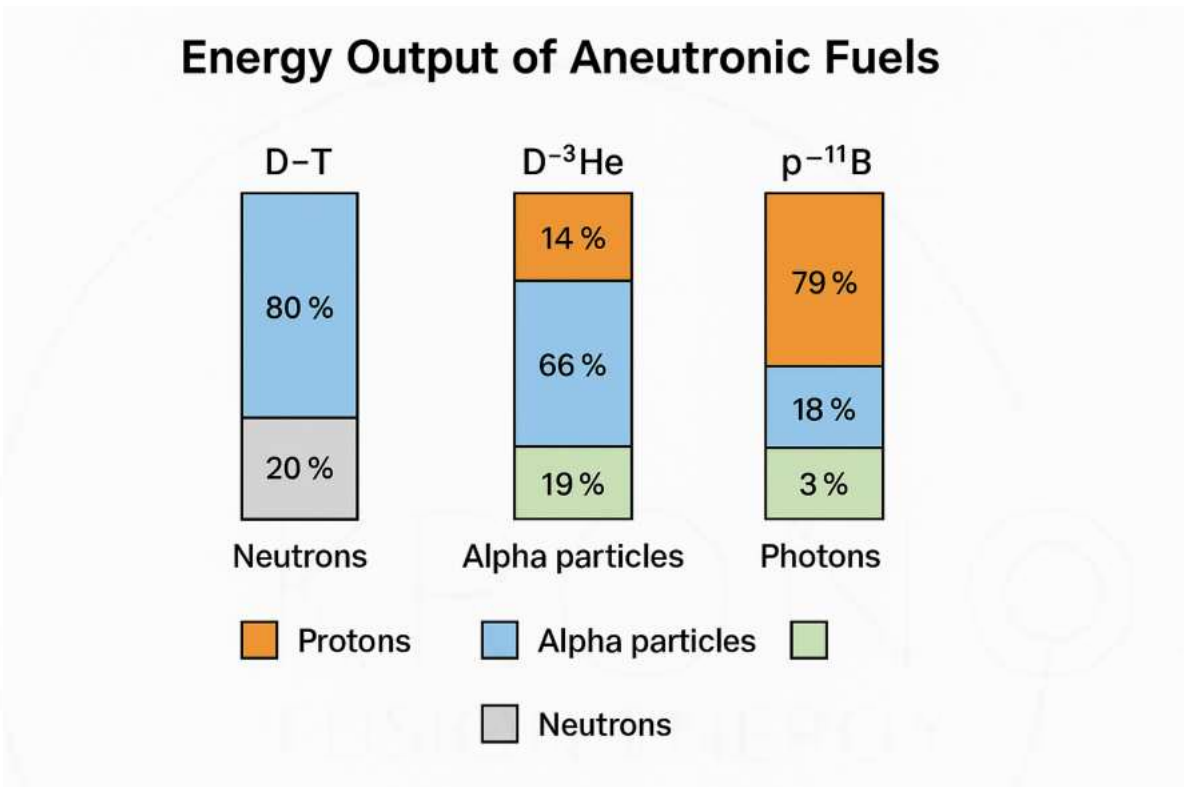

Figure 2: Comparative energy distribution for D–T, D–^{3}He, and p–^{11}B fuels. Unlike D–T, which emits most of its energy as neutrons, aneutronic fuels release ≥95 % of their energy in charged particles and photons—making them highly compatible with direct energy conversion (DEC) systems.

yielding a **total of ~18.3 MeV** per reaction[12]. The kinetic energy is split between a 3.6 MeV **alpha particle** (helium-4 nucleus) and a **14.7 MeV proton**, both charged. Importantly, this primary branch emits *no neutrons*. However, in a plasma, deuteron–deuteron side reactions can occur in parallel, producing a small fraction of D–D fusion products (e.g. a 2.5 MeV neutron from D+D→T+n)[12]. In a D–^{3}He reactor with deuterium "lean" mixture, these side reactions are minimal – on the order of a few percent of the total power[13]. Thus, practically **~95–98%** of the fusion energy of a D–^{3}He plasma can be released as charged particles or photons, making it highly amenable to direct conversion. The charged proton at 14.7 MeV is especially significant: as a light, singly-charged ion it carries most of the energy and can be manipulated by electric and magnetic fields for direct energy extraction.

The **p–^{11}B fusion** reaction is often cited as "truely aneutronic." It proceeds via a chain yielding three alpha particles and no neutrons:

$$p + {}^{11}B \rightarrow 3\,{}^{4}He + 8.7\ MeV.$$

Each reaction liberates **8.7 MeV** of energy carried by **three alpha particles** (helium-4) in the final state[14]. In reality, the mechanism is $p + {}^{11}B \rightarrow {}^{12}C \rightarrow$ *3 α, involving intermediate excited carbon-12. The three alphas share the energy (on average ~2.9 MeV each). Aside from very rare side channels, no neutrons are produced. All fusion energy is initially in charged particles. This makes p–^{11}B* in principle *ideal for DEC. The challenge, however, is that p–^{11}B has a higher ignition temperature (~123 keV, nearly 10× that of D–T) and suffers strong radiation losses at thermal equilibrium[15][16]. Thus, a reactor burning p–^{11}B likely requires non-Maxwellian plasma schemes or pulsed inertial approaches. Assuming such schemes can produce energetic alphas, essentially 100% of the fusion energy is in charged kinetic form** (with only prompt X-rays from plasma bremsstrahlung/synchrotron as losses). That charged output can be directly converted if efficiently collected.

Output distribution by particle type – implications for DEC: In both fuel cases, the majority of energy can be funneled into charged reaction products. D–^{3}He yields a fast proton which, if allowed to escape the plasma or reach a collector, carries ~80% of the reaction energy[12]. The helium ions (whether 3.6 MeV alphas from D–^{3}He or 2–3 MeV alphas from p–^{11}B) carry the remainder. These charged ions can be directly decelerated to produce electric current. By contrast, conventional D–T fusion emits 14.1 MeV neutrons carrying 80% of energy which **cannot** be directly converted to electricity (neutron energy must become heat in a blanket) – a fundamental limit avoided by aneutronic fuels[1]. Any electromagnetic radiation produced in aneutronic plasmas (e.g. bremsstrahlung X-rays from electron-ion collisions, or synchrotron radiation from hot electrons in a magnetic field) can in principle also be directly converted using appropriate photovoltaic or antenna-based devices. In D–^{3}He, a high plasma temperature (~50–100 keV) is needed, which does lead to significant bremsstrahlung and cyclotron radiation – Apollo-L3 predicted roughly 50% of D–^{3}He plasma power as synchrotron and ~27% as bremsstrahlung[1]. These photons, though initially a loss mechanism, are high-frequency energy that DEC can target. In summary, **aneutronic reactors output energy in three forms**: fast charged ions (α or p), electromagnetic radiation (X-ray, UV, etc.), and a very small neutron fraction. All but the neutron fraction are directly convertible by an appropriate modality. The following sections explore specific converter technologies matched to each form of energy output.

Modal DEC Architectures

A. Electrostatic Direct Collectors – These devices convert the kinetic energy of charged fusion products (ions or electrons) directly into a DC electric potential. The classical design is a **multi-stage grid decelerator**, often exemplified by the **"Venetian blind"** converter proposed by R. F. Post and developed by Moir and Barr in the 1970s[17][18]. In such a system, divergent charged particles from the plasma first pass through a magnetic expander or nozzle that collimates and guides them out of the confinement region[19][20]. Electrons are then diverted (e.g. by suppressor grids or magnetic fields) so that mainly positive ions stream into an array of biased collector grids. The **ribbon-like grids**, angled like a Venetian blind, present a series of increasing electric potentials that the ions must climb[21]. High-energy ions penetrate deeper before being turned around by an opposing field, whereas lower-energy ions turn back earlier, thus ions "sort" themselves by energy and are collected on electrodes of corresponding voltage[22]. Each stage thus extracts a portion of the kinetic energy by decelerating the ions and charging the electrode to a high voltage. By the time ions reach the final stage, ideally they are slowed to near zero energy

and collected. The extracted charge flows through an external circuit, delivering electric power at high DC voltage (on the order of tens to hundreds of kV per stage).

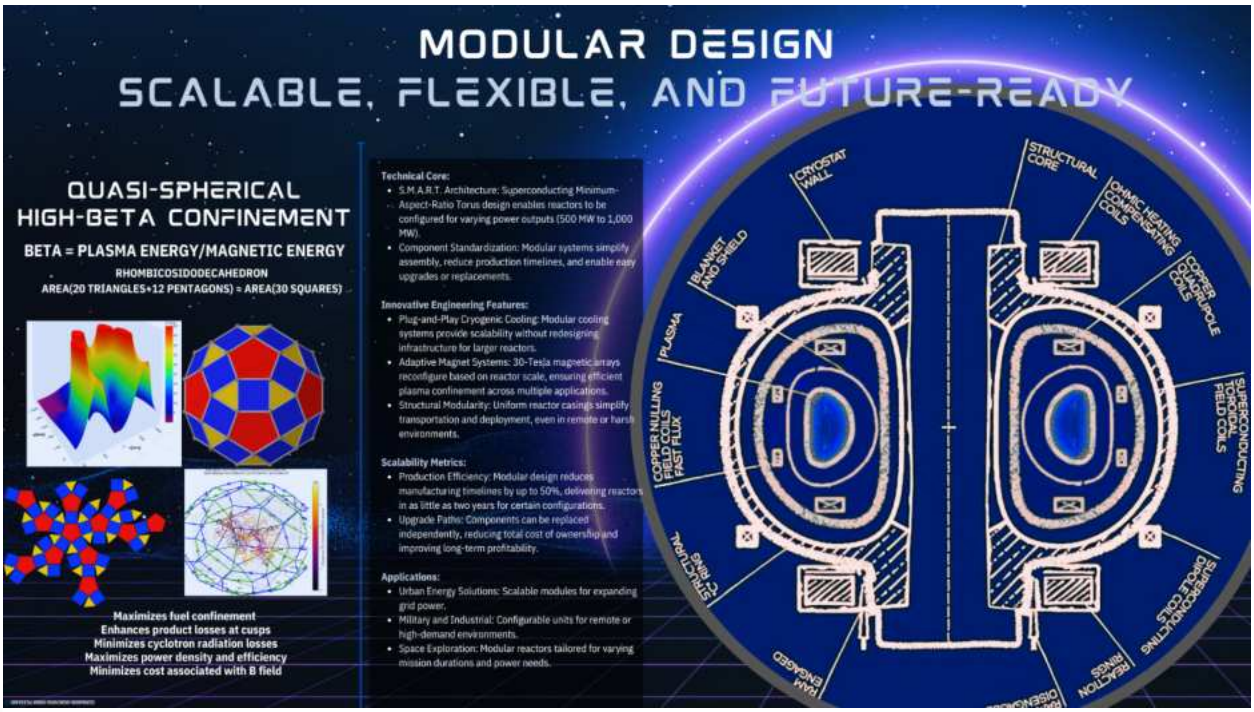

Figure 3: Venetian-blind electrostatic direct energy converter. High-energy ions are decelerated in multiple field stages, converting kinetic energy to voltage while preserving plasma space-charge neutrality.

Multi-stage electrostatic collectors can be remarkably efficient in theory – adding more stages (more finely segmented deceleration) approaches 100% energy recovery[23][24]. In practice, **space-charge limits** and beam neutralization are key constraints. As positive ions slow down, their density tends to build up and repel following ions unless sufficient electrons are present to neutralize the space charge. In mirror-based direct converters, the escaping plasma was quasi-neutral over small Debye lengths, but in larger systems maintaining neutrality is non-trivial[25]. Rosenbluth noted that holding the plasma charge quasi-neutral within a few centimeters (typical Debye length for 100 keV plasma) would be difficult, risking ion beam blow-out[8]. Engineering solutions include introducing low-energy electrons or neutrals to neutralize the ion beam between stages, or operating at low current densities per stage. The Venetian blind geometry inherently helps space-charge handling by allowing most ions to pass through each grid (which is largely transparent to incoming ions but opaque to reflected ions)[17][26]. This geometry demonstrated excellent space-charge tolerance down to ~10 keV ion energy in experiments[3].

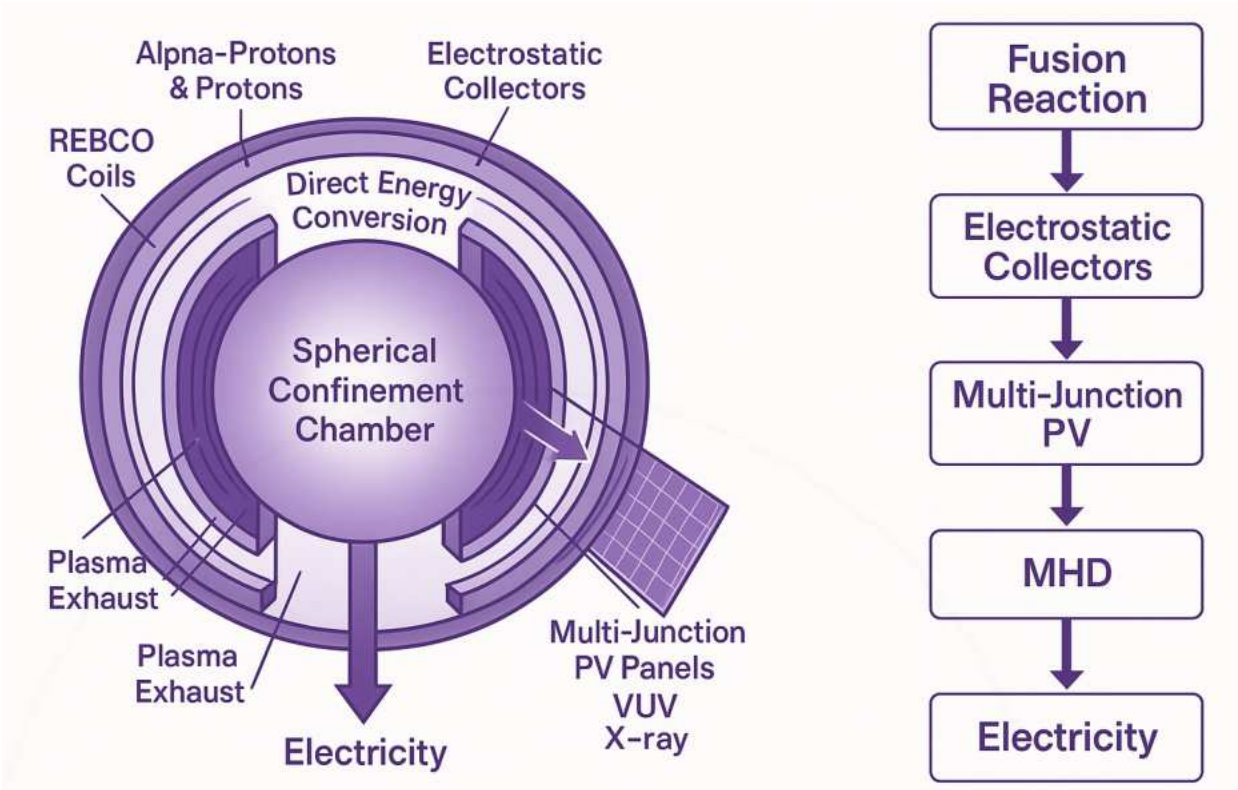

Figure 4: Modal direct energy conversion (DEC) pathways in the S.M.A.R.T. architecture. Charged fusion products are decelerated electrostatically or harvested in MHD channels, while photons are captured via thermionic or radiovoltaic converters. These front-end systems capture the majority of fusion output energy before residual heat enters a Brayton bottoming cycle.

Efficiency and performance: Experimental benchmarks are promising. Moir *et al.* achieved **65% peak energy capture** (50% minimum) with a two-stage electrostatic collector on ion beams[6]. In the TMX tandem mirror test, a single-stage direct collector recovered 79 W out of 138 W of plasma kinetic power (after subtracting suppressor grid power, net **48% efficiency**)[27]. A two-stage Venetian-blind unit was projected to reach ~60% efficiency on that device[28]. Theoretically, in D–T plasma exhaust (~100–150 keV ions), venetian-blind converters could operate ~60% efficient under realistic load, with an upper limit ~70% if engineering constraints are ignored[29]. For aneutronic fuels, which produce even higher energy ions (e.g. 3 MeV alphas, 15 MeV protons), multi-stage deceleration could in principle capture an even larger fraction of energy. The high ion charge-to-mass ratio (e.g. proton vs. alpha) might require tuning – for example, a magnetic cusp can be used upstream to separate heavier alphas from protons, allowing optimized deceleration voltages for each species[30][31]. Energy selectivity is inherent to the multi-grid design: each grid stage only captures ions above a certain energy (those that penetrate that far before turning). This minimizes interception of lower-energy ions at high-voltage stages, thus reducing down-scattering losses.

Despite these advantages, challenges remain. **Grid interception and heat loads:** Some ions will hit the grids at high energy, depositing heat. Grid materials must withstand these loads and neutron/ion damage (in aneutronic reactors, neutron flux is low but not zero for D–^{3}He). **High voltage insulation:** Multi-stage converters operate at hundreds of kV potential differences in a plasma environment – designing insulators or electrostatic configurations that prevent arcing and minimize electron leakage is non-trivial[32][33]. **Conversion of power to usable form:** The direct

collector outputs DC at high voltage. This can either feed directly into a high-voltage DC bus or be inverted to AC. Prior design proposals include converting the DC output to AC via solid-state inverters for coupling to the grid[34]. Nonetheless, electrostatic DEC remains one of the most mature and efficient approaches for charged fusion products. Advanced variants, like **traveling-wave direct converters (TWDEC)** or **inverse cyclotron converters (ICC)**, have been proposed for aneutronic plasmas: these use RF fields or cyclotron resonance to decelerate ions gradually. For example, a TWDEC system was predicted to achieve ~70% efficiency converting 1–3 MeV protons in a D–^{3}He reactor[35][36]. Such concepts aim to address space-charge by using oscillating fields and continuous deceleration rather than discrete grids. While still conceptual, they reinforce that electrostatic/electrodynamic conversion could capture on the order of **two-thirds or more** of fusion charged-particle energy with proper engineering.

B. Magnetohydrodynamic (MHD) Generation – MHD conversion is a form of direct energy extraction from a moving conductive fluid (plasma or ionized gas) via electromagnetic induction. In a fusion context, one can envision using the **plasma exhaust stream** (for instance, a jet of charged plasma from a divertor or an expanding FRC plasmoid) as the "working fluid" in an MHD generator. The principle is governed by the Lorentz force: as charged particles move through a magnetic field **B** with velocity **v**, they experience a force **q(v×B)** that drives positive and negative charges in opposite directions. By placing electrodes perpendicular to the flow and magnetic field, an electric current can be drawn from this charge separation. In essence, the plasma's kinetic energy is directly converted to electrical energy through electromagnetic coupling, generating a DC current in the external circuit[37][38]. The open-circuit induced electric field is **E = v×B**, pointing from the "+" electrode (collecting positive ions/electrons deflected one way) to the "–" electrode (collecting the opposite charge carriers). The power density of an ideal MHD generator can be estimated as $P \sim \sigma(v \times B)^2$, where σ is the electrical conductivity of the plasma. In a high-temperature plasma with sufficient density, σ can be large, and strong fields and fast flows yield substantial power.

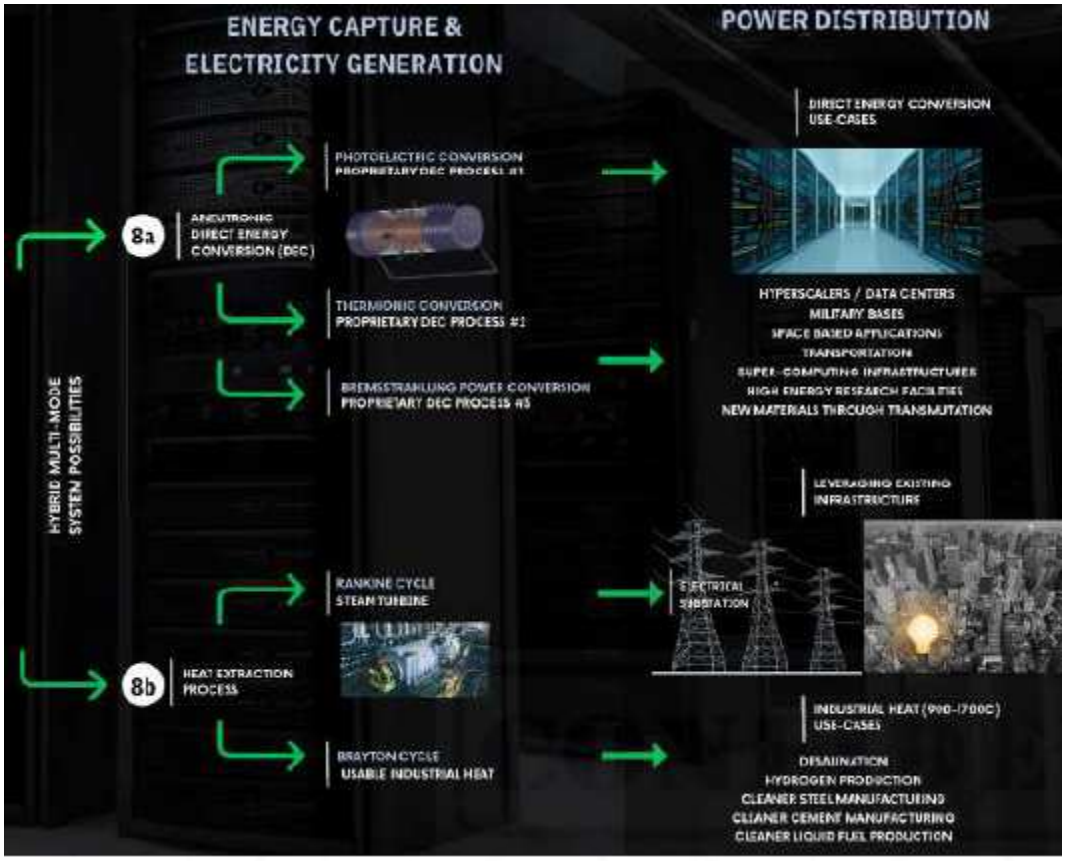

Figure 5: Diagram of energy capture and electricity generation leading to power distribution. On the left, aneutronic direct energy conversion (DEC) processes include photoelectric, thermionic, and bremsstrahlung recovery, alongside heat-extraction pathways via Brayton and Rankine cycles.

For **charged fusion exhaust**, one could design a divergent magnetic nozzle that guides the plasma out of the tokamak and into an MHD channel. Both ions and electrons would be deflected – e.g. ions pushed toward the channel floor, electrons toward the ceiling – driving a current between those walls[38]. In practice, some separation of charges might be needed (for example, using a transversal magnetic field (Faraday configuration) or Hall configuration). One advantage is that unlike electrostatic grids, the MHD process is continuous and has no physical grids intercepting the flow; the plasma itself acts as the conductor. Additionally, **both ions and electrons contribute** to current (they deflect oppositely, effectively doubling the charge separation)[38]. MHD generators have been studied extensively in the context of topping cycles for coal-fired plants (e.g. seeding a combustion plasma with potassium to raise conductivity and extracting ~10% additional efficiency). In those cases, temperatures ~2500 K achieved ~20–30% conversion efficiency in practice, and combined cycle efficiencies reached ~50–60% (MHD + steam)[39][40].

In a fusion scenario, the plasma exhaust would be much hotter (10–100 eV, i.e. 100,000–1,000,000 K). However, a fully ionized plasma at such temperatures is also fully conductive and tends to *short itself out* unless appropriately configured (pure plasma might just carry internal currents rather than driving external load). Thus, MHD fusion schemes might introduce a *tampor* or carrier gas. One concept is to mix the charged fusion products into a stream of helium or CO_2 working gas that is pre-heated by the fusion core; this gas is seeded and run through an MHD generator as in conventional plants[41]. That blurs the line with a thermal cycle. A more direct concept is **open-cycle plasma MHD**: use a plasma jet directly. For example, in mirror-based

reactors, the end-loss plasma could directly expand through an MHD duct. The key parameters are the flow velocity (set by plasma exhaust speed, potentially 10^5–10^6 m/s for MeV ions), the magnetic field (several Tesla), and the plasma conductivity (dependent on density and temperature). If optimized, substantial voltages (order kilovolts) can be generated across the channel.

One illustrative equation is the MHD generator's efficiency in an ideal Faraday generator: $\eta_{\mathrm{MHD}} = vBL/(vBL + R)$, where L is the span between electrodes and R represents losses (due to finite conductivity and cycle extraction). This indicates efficient conversion requires high flow velocity v, strong B, and good conductivity. In fusion, v is fixed by particle energy (e.g. 15 MeV protons move ~5–6% of speed of light). Achieving directed flow might require converting isotropic particle motion into bulk flow (via magnetic expansion nozzles, as Post described[19]). MHD also generates drag on the plasma – it literally decelerates the flow as energy is extracted. This is desirable (it slows escaping plasma, reducing exhaust losses), but the plasma must remain stable while decelerating.

Although experimental data on plasma MHD generation in fusion devices is scarce, analogous *plasma thruster* research and space propulsion studies suggest it is feasible. A Defense IA analysis noted that on a spacecraft, an MHD generator could **directly generate power from fusion exhaust ions**, with ions and electrons deflected to opposite electrodes by a magnetic field[38]. This would simultaneously produce thrust and electricity in an aneutronic rocket concept. The **v×B electric field** has been demonstrated in plasma wind tunnels and Hall thrusters, lending credence to the concept. If realized, MHD could convert a significant fraction (perhaps 20–40%) of the plasma kinetic energy to electricity before the plasma is either diverted to a thrust or dumped as heat. MHD channels would likely be used in tandem with electrostatic collectors – e.g. an initial MHD stage for bulk deceleration followed by fine electrostatic stages. One must also consider electrode arcing, plasma instabilities induced by MHD drag, and the need for insulating or segmented electrodes to avoid short-circuiting via Hall currents. Research is ongoing on advanced MHD materials and designs (e.g. using high-temperature ceramics for electrodes and superconducting magnets for high B fields). In summary, **MHD channels offer a robust, moving-part-free means** to extract electrical power from a streaming plasma, directly proportional to plasma flow power. They complement electrostatic converters by handling very high-power-density flows and could form an effective first-stage decelerator for charged fusion exhaust.

C. Thermionic Converters – Thermionic energy conversion (TEC) is a **solid-state direct conversion** method that uses heat to drive electrons across a vacuum or vapor gap, generating electricity. In a fusion reactor, abundant **thermal radiation** (X-rays, bremsstrahlung) and hot plasma-facing components can heat surfaces to extremely high temperatures (1000–1700 °C). Thermionic converters can be placed at these hot surfaces to directly produce DC power. A typical thermionic converter consists of an **emitter electrode** (hot cathode) and a **collector electrode** (cooler anode) separated by a small gap. The emitter is heated to a high temperature such that electrons overcome its work function and boil off (thermionic emission). These electrons flow to the cooler collector, creating a current. The output voltage is roughly the difference in work functions (emitter vs collector) minus any space-charge voltage drop.

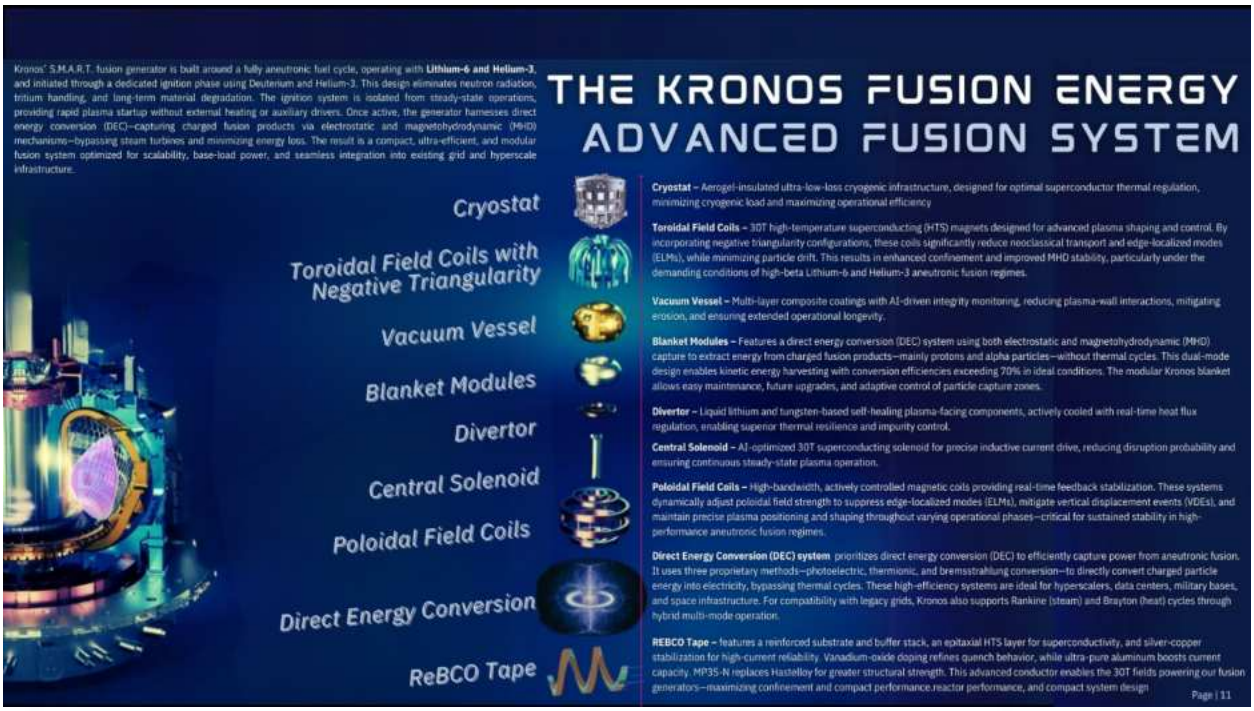

Figure 6: Conceptual thermionic converter cell. Electrons are emitted from a hot cathode surface and collected at a cooler anode, producing direct current at efficiencies up to ~20 % in high-temperature regimes.

Key to high performance is a low work function emitter material that can survive high temperature. **Lanthanum hexaboride (LaB_6)** and **cerium hexaboride (CeB_6)** are among the best-known advanced emitters. They have work functions on the order of $\varphi \approx 2.5$–2.7 eV (for LaB_6) and ~2.7–2.9 eV (CeB_6), significantly lower than plain metals[42][43]. They are also refractory, with LaB_6 usable up to ~1700 K (and melting above 2200 K). By comparison, traditional tungsten emitters have $\varphi \approx 4.5$ eV and require cesium vapor to reduce the barrier (cesiated tungsten can achieve an effective $\varphi \approx 1.5$–2 eV but only under carefully controlled conditions)[44][45]. In fact, studies showed that with cesium adsorption, LaB_6's work function could be lowered to ~1.3 eV[44], potentially yielding very high efficiencies – but cesium introduction brings complexity. Even without cesium, LaB_6 at 1700 K has an effective $\varphi \approx 2.5$ eV, enough to emit ~10 A/cm² of electron current[46]. CeB_6 has similar emission properties but slightly higher vapor pressure (thus a bit less thermally robust)[45].

The thermionic emission current density is given by the **Richardson–Dushman equation**:

$$J = AT^2 \exp\left(-\frac{\Phi}{kT}\right),$$

where T is emitter temperature, Φ the work function, and A the Richardson constant (~120 A/cm²·K² for an ideal surface)[47][48]. This exponential sensitivity to Φ and T means that high T and low Φ are crucial for strong electron output. For example, an LaB_6 emitter at 1800 K ($\Phi \approx 2.5$ eV) can output on the order of 10 A/cm²[46]. The converter's efficiency depends on how much of the heat flux is converted to electrical power versus lost as thermal radiation or conduction

to the collector. In practice, **20%–30% efficiency** has been achieved in specialized thermionic converters at high temperatures. NASA studies in the 1970s indicated that a converter with a 1800 K emitter and ~0.7 eV effective work function difference could reach ~30% efficiency[44][49], whereas at 1600 K it might be ~15%[50]. Indeed, experimental cesiated tungsten converters demonstrated ~15–20% efficiency in terrestrial tests, and the Soviet TOPAZ space nuclear reactors (thermionic fission reactors) operated at ~10% efficiency with multi-cell thermionic stacks. With advanced materials like LaB_6, even higher efficiencies are expected due to lower φ. For instance, a projection was **31% efficiency at 1800 K** for a LaB_6-based converter with optimized gap and collector[44].

Several engineering aspects are notable: **Interelectrode plasma or vacuum:** Thermionic converters often introduce a low-pressure cesium vapor in the interelectrode gap to mitigate space-charge (the positive ions neutralize electron space-charge, allowing more current at lower voltage drop). In a fusion environment, one might instead use the inherent plasma or a controlled gas in the gap. For simplicity in space, **vacuum thermionics** can be used, but they suffer space-charge limits that require very small gaps (~0.1 mm or less) or nanostructured geometry to allow high current density. **Emitter/collector geometry:** Planar diodes are common in lab tests, but cylindrical or concentric geometries can pack more surface area. In a tokamak first wall application, one could envision emitter panels on the inner wall, with cooled collectors a short distance away. The emitter could be the plasma-facing material itself if it can be made of or coated with LaB_6 (though LaB_6 is not very erosion-resistant, so more likely a protected thermionic module just behind the first wall). **Cooling and heat management:** The collector must be cooled (often ~600–800 K in designs, to maintain a good temperature difference). The waste heat from the collector can either be radiated or passed to a secondary cycle (e.g., a Brayton cycle). In hybrid systems, thermionic converters act as a topping cycle, converting the hottest portion of heat directly to electricity, and a Brayton or Rankine cycle converts the remainder. This two-tier conversion was shown to potentially exceed 50% combined efficiency[51].

Thermionic converters are attractive for aneutronic fusion especially in **space or mobile applications**: they are all-solid-state, with no moving parts (aside from maybe a liquid-metal coolant loop), and can operate in vacuum (space provides an ideal environment for maintaining the necessary vacuum or controlled vapor). Materials like LaB_6 cathodes have flight heritage in satellites (as electron sources) and can operate for thousands of hours. Challenges include material stability (LaB_6 will slowly evaporate at >1600 K; it needs to last for years)[45], and maintaining the necessary vacuum gap in a high-radiation environment (outgassing, sputtering, etc., could degrade the gap or deposit impurities on electrodes). Nonetheless, research continues into **nano-structured thermionics** (which use nanogaps or emitter arrays to enhance current and reduce space-charge limits) and **thermionic-photovoltaic hybrids** (where the emitter also emits tailored spectra for a photovoltaic cell). In the context of DEC, thermionics directly address the **photon and thermal energy** portion: for example, the soft X-ray and extreme UV power from a D–^{3}He plasma will heat surfaces to glowing, and a thermionic converter can turn that heat to electricity at the source, before it is diluted. Thus, thermionic converters can handle the part of fusion output that is inevitably thermal (transport losses, wall heating, etc.), improving overall efficiency and reducing radiator load (especially in space systems).

D. Photoelectric / Radiovoltaic Conversion – In aneutronic fusion, a significant fraction of power may be emitted as **bremsstrahlung X-rays and synchrotron radiation** (especially for D–^{3}He at high plasma temperature)[1]. Directly converting this high-frequency photon energy is challenging

but under active investigation. Two approaches exist: (1) **Photoelectric converters (X-ray photo-cells)** consisting of stacked high-Z foil layers that absorb X-ray photons and emit photoelectrons, which are then collected as current; and (2) **Radiovoltaic cells**, essentially photovoltaic devices optimized for ionizing radiation (deep ultraviolet, X or γ).

The **multi-foil photoelectric approach** was conceptually described in early aneutronic fusion literature[52]. X-ray photons (which have energy quanta from a few keV up to possibly MeV range in p–^{11}B plasmas) can penetrate significant thicknesses of material. When an X-ray passes through a metal foil, it can eject an electron (photoelectric effect). If many foils are stacked, an X-ray will statistically interact in one of them, yielding a cascade of electrons. By applying an electric field across the stack (or biasing alternate foils), those liberated electrons can be drawn off as current. Essentially, the device acts like a giant stacked solar cell for X-rays, using many layers to capture the photons that would otherwise pass through. In practice, **hundreds or thousands of layers** might be needed because X-rays are so penetrating[53]. Each layer absorbs only a few percent, so a thick stack ensures most photons eventually produce an electron. The photoelectrons are low-energy (~keV) and can be collected with modest bias. The overall efficiency of such a converter is currently limited – estimated in single-digit percentages – because of recombination and the fact that each X-ray yields at best one electron (so quantum efficiency is limited, and a lot of photon energy may go into wasted heat in the foils)[52]. However, it is a straightforward, robust approach (no semiconductor bandgap limitations). Los Alamos experiments in the 1980s on "X-ray diodes" showed it was possible to rectify hard X-rays to DC, though efficiency was low.

Modern research has turned to **semiconductor radiovoltaics**. These are akin to betavoltaic or gammavoltaic nuclear batteries: a semiconductor p-n junction that directly produces electricity when hit by high-energy radiation. Traditional solar cells (Si, GaAs) are optimized for visible light (~1–3 eV photons). An X-ray or gamma photon (10 keV–1 MeV) will pass through such a cell, depositing energy as it goes – potentially creating thousands of electron-hole pairs, but also damaging the lattice. The key is to use materials with wide bandgaps and high radiation tolerance, and perhaps to incorporate luminescent or down-conversion layers. **Gallium arsenide (GaAs)** and related III-V semiconductors have been explored because they have direct bandgaps that can be tuned (GaAs bandgap 1.43 eV, but GaN is 3.4 eV, SiC ~3 eV, etc.) and they have better radiation hardness than silicon. Stacking multiple junctions (e.g. perovskite top cell with bandgap ~2 eV, GaAs middle, Ge bottom) could cover a broad spectrum so that an X-ray deposits energy and a portion of it gets converted to electrical output. However, for very high energy photons, typically one needs an intermediate step: a **scintillator** or luminescent material that absorbs the X-ray and re-emits lower-energy photons (visible/UV) which a photovoltaic can then capture. This is essentially a radioluminescent approach feeding a solar cell, and its efficiency is limited by the Stokes loss (high-energy photon down-converted to many lower-energy ones).

A significant development is the recent DARPA **"Rads-to-Watts"** program[54][55]. It is funding teams to create **high-power radiovoltaic cells** that can directly handle intense radiation flux. The aim is to bypass the "thermal middleman" even for nuclear reactors – exactly the aneutronic fusion philosophy. Over 60 material concepts were proposed, leveraging new materials with high carrier mobility and strong absorption of high-energy photons[56]. Some approaches likely involve advanced perovskite layers (perovskites can be engineered to different bandgaps and are relatively easy to process in thick layers) and diamond or GaN diodes (diamond has an ultra-wide bandgap ~5.5 eV and exceptional radiation hardness, making it a candidate for gamma-voltaic cells[57]). One recent result, for instance, is a **diamond Schottky diode** that converts gamma radiation to

current with promising efficiency[57]. Likewise, GaN p-n junctions have been made into betavoltaic devices for beta emissions with ~5% efficiency in lab tests[58]. While these efficiencies are modest, the field is rapidly advancing with new nanostructures and conductive oxide interfaces to improve charge collection under high radiation flux.

For fusion DEC, a radiovoltaic system could take the form of **panel arrays lining the vacuum vessel** behind the first wall. They would absorb bremsstrahlung X-rays (which are mostly in the ~10–500 keV range for 50 keV electrons) and possibly UV from synchrotron. A layered design might have a high-Z backing to absorb and re-radiate X-rays as lower-energy photons into an adjacent photovoltaic layer. Another concept is a rectenna for electromagnetic radiation: Apollo-L3's design in fact used **microwave rectennas** to capture long-wavelength synchrotron radiation at 80% efficiency[59][60]. For shorter wavelengths (far-UV to soft X-ray), a nano-antenna array with high-speed diodes could in theory rectify these oscillating fields as well – an approach being studied in the context of thermal infrared rectennas for energy harvesting. Such optical rectennas would need THz to PHz frequency diodes, which are extremely challenging (quantum tunneling diodes or plasmonic structures might be required). Given these challenges, the more pragmatic short-term approach is to use **multilayer semiconductor converters**. Efficiency estimates for near-term radiovoltaics are in the 10–20% range for very high radiation fluxes (DARPA's program aims to outperform radioisotope thermoelectric generators, which are ~7%, and approach >20%). Even at 10% efficiency, capturing 20% of a fusion reactor's power that is radiative would add 2% net efficiency – not huge, but every bit counts, and it reduces heating of structures. In the best case, pairing radiovoltaics with thermionics can double-dip the photon energy: e.g., capture high-energy photons electrically, and whatever is not captured just heats a thermionic emitter to produce additional power. This synergistic approach could significantly raise the conversion of photon power.

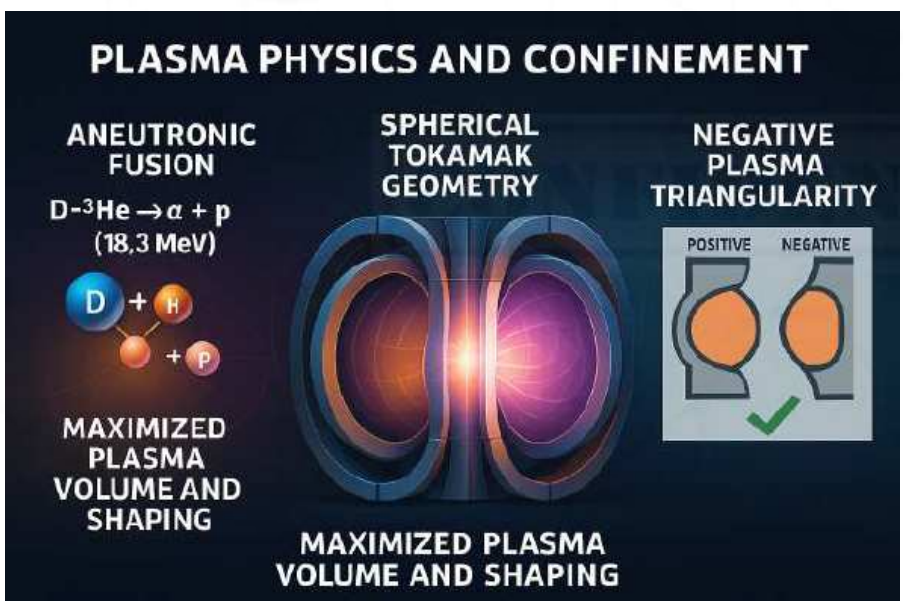

Figure 7: Multilayer radiovoltaic stack for direct photon-to-electron conversion. High-energy photons are absorbed in alternating semiconductor layers to produce current, with unconverted energy captured by a thermal Brayton loop.

In summary, photoelectric and radiovoltaic DEC systems target the **photon output** of aneutronic plasmas. They are less mature than charged-particle converters but hold the key to harnessing the sizable X-ray power that otherwise would be lost as heat. Ongoing advances in materials (GaAs,

GaN, perovskites, diamond) and nanostructures (quantum dots, luminescent concentrators) are steadily improving the prospects for turning "waste" radiation into electricity in future fusion reactors[61][62].

Hybrid System Integration

No single conversion method will capture *all* the energy in a fusion reactor. A logical strategy is a **hybrid DEC + thermal cycle**, where direct conversion handles the high-grade kinetic and photon energy, and a secondary Brayton cycle converts the remaining thermal energy. This is analogous to combined-cycle power plants today (Brayton gas turbine topping a Rankine steam bottoming cycle), but here the "topping cycle" is a set of direct converters. The hybrid integration is especially pertinent for aneutronic reactors: even if, say, 70% of fusion energy is ideally extractable via DEC, the remaining ~30% (as low-energy heat, unconverted radiation, etc.) can still be fed to a high-temperature turbine to push efficiency higher.

Fusion Energy Conversion Pathways, Technologies, and Efficiencies

Fusion Energy Channel	Conversion Technology	Efficiency (η)	TRL
Charged particle kinetic (α particles, protons)	Electrostatic direct collectors; MHD induction generators	~60–80% (direct conversion)	3–4 (experimental)
Photon radiation (bremsstrahlung X-rays, synchrotron)	Multi-layer photoelectric converters (GaAs/perovskite PV); thermal absorption (Brayton gas turbine)	PV: ~10–20%; Brayton ~45–50% (combined ~50% of photon energy)	PV: 2–3 (concept); Brayton: 8–9 (industrial)
Thermal residual (exhaust heat, structures)	Secondary Rankine cycle (steam turbine generator)	~30–35%	9 (mature)

Table 1: Fusion energy conversion pathways, technologies, and efficiencies. Charged particle kinetic energy (α particles, protons) is captured via electrostatic direct collectors and MHD induction generators with ~60–80 % efficiency at TRL 3–4. Photon radiation (bremsstrahlung, synchrotron) can be converted using photoelectric PV (~10–20 %) or thermal Brayton cycles (~45–50 %), corresponding to TRL 2–3 for PV and 8–9 for Brayton. Residual thermal energy is recovered through mature Rankine steam turbine cycles (~30–35 % efficiency, TRL 9).

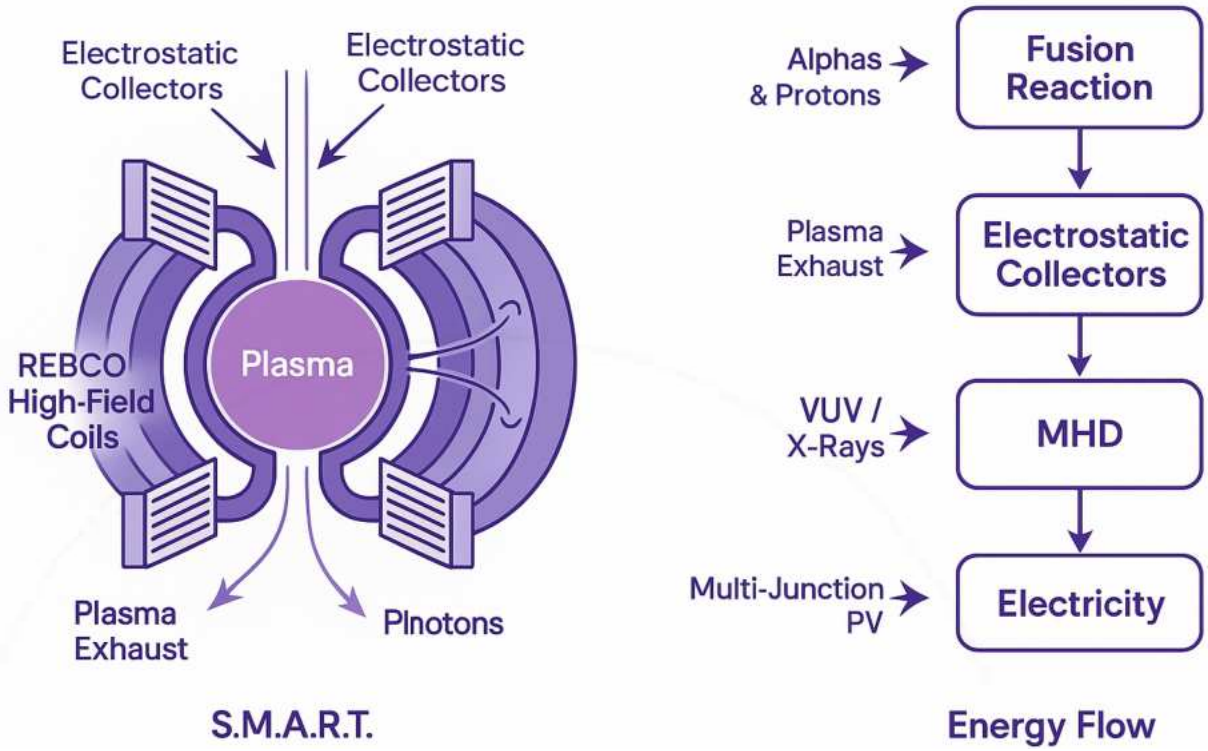

Figure 8: End-to-end energy flow in the S.M.A.R.T. reactor. Plasma energy is split between charged-particle direct conversion, high-efficiency Brayton/sCO_2 thermal cycles, and minor Rankine recovery, yielding a combined net electrical efficiency ≥70 %.

High-Temperature Brayton Cycles: The leading thermal conversion option for advanced reactors is a closed Brayton cycle using either supercritical CO_2 or helium as the working fluid. These fluids can operate at turbine inlet temperatures far above what steam can tolerate. In fusion designs with refractory structural materials (e.g. W or SiC composites), coolant outlet temperatures of 800–1000 °C have been considered[5]. Some advanced concepts (using carbon-based materials or molten salts) even target 1300–1600 °C. A helium Brayton cycle at 1400 °C (1673 K) could achieve thermal efficiency on the order of 55–60%, considering Carnot efficiency $1 - T_{\text{cold}}/T_{\text{hot}}$ would be ~80% and modern turbomachinery can reach 70–75% of Carnot[63][64]. Supercritical CO_2 cycles are very attractive at intermediate temperatures (450–750 °C) due to their compactness and good efficiency (~45–50% at 750 °C)[65][66]. They tend to hit materials limits ~700 °C unless using very advanced alloys. Helium can go hotter (it's inert and single-phase), limited by turbine blade materials (Ni-based superalloys top out ~1100 °C, ceramics like Si_3N_4 or Mo-based alloys might extend to 1300+ °C). The U.S. DOE and EU programs (e.g. **STEP demo** in the US, and studies for DEMO fusion plant) are actively developing sCO_2 Brayton technology for nuclear applications[67][68]. A European study comparing cycles for a fusion DEMO found sCO_2 recompression cycles could rival helium in net efficiency while being much smaller in size[65][66].

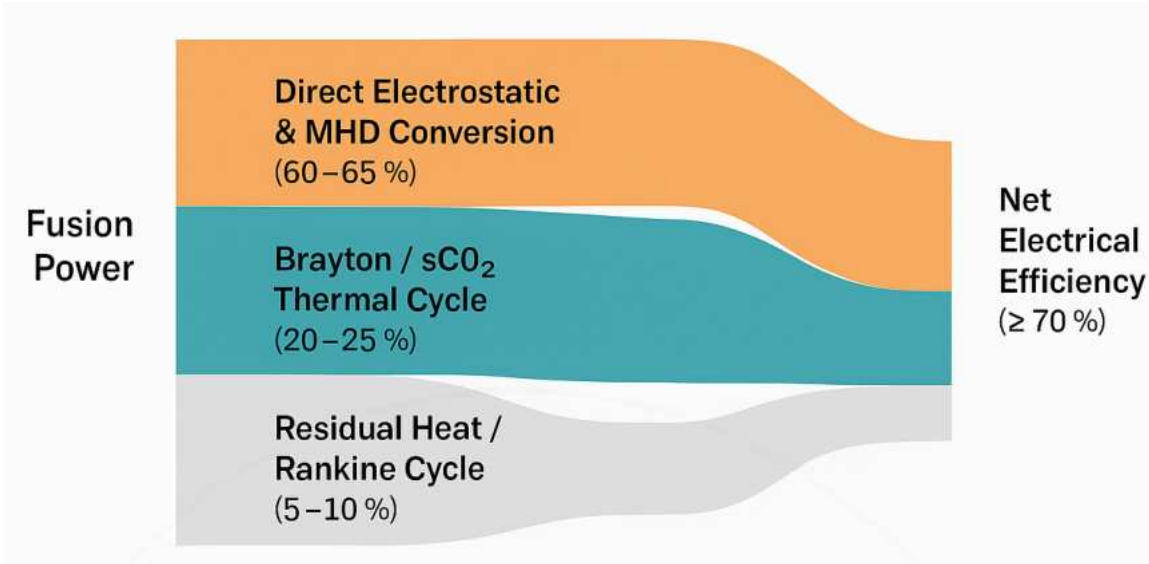

Figure 9: Sankey diagram of hybrid energy recovery in the S.M.A.R.T. architecture. Multi-modal direct conversion captures 60–65 % of charged-particle energy, while a Brayton/sCO_2 cycle and residual Rankine cycle recover additional thermal power, achieving ≥70 % net electrical efficiency.

For a hybrid aneutronic reactor, one could employ **DEC as a front-end** that directly outputs DC power from charged particles and certain photons, and a **Brayton cycle as the back-end** to convert the remaining thermalized energy. The integration can be understood with an energy balance. Let η_DEC be the fraction of total fusion energy converted by direct means, and (1–η_DEC) is then the fraction remaining as heat. If the Brayton cycle converts that heat with efficiency η_B, the **overall efficiency** is:

$$\eta_{\text{total}} = \eta_{\text{DEC}} + (1 - \eta_{\text{DEC}})\,\eta_B.$$

This formula assumes the DEC and thermal paths operate in parallel on the partitioned energy. For example, Apollo-L3 had η_DEC ≈ 0.30 (directly converting synchrotron) and a thermal cycle of η_B ≈ 0.44, yielding η_total ≈ 0.30 + 0.70*0.44 = 0.61 (61%) if both ran ideally – in practice Apollo-L3 achieved ~47% due to not converting all synchrotron and using a lower-temperature cycle[9][10]. Let's consider a more optimized aneutronic plant: suppose 60% of fusion energy can go through DEC (between fast ions and some photons) at high efficiency, and 40% ends up as heat. If DEC captures, say, 50% of the total fusion energy (η_DEC=0.5 as electricity), then the remaining 50% goes to heat; a Brayton cycle at 50% efficiency on that yields another 25%. The total would be ~75% – extraordinarily high for any power plant. If one pushes η_DEC to 0.7, then (1–η)=0.3 remains; Brayton at 50% would add 0.15, totaling 85%*. This begins to approach the fundamental limits (Carnot limit from a ~1 keV plasma temperature to ambient is extremely high, but practical limits come in elsewhere). While 85% may be aspirational, studies have indeed suggested overall efficiencies >70% could be possible for D–^{3}He reactors using combined direct and thermal conversion[51]. By contrast, a "Rankine-only" D–T reactor like ARIES-I managed ~39%[69], and even using an advanced Brayton at 1000 °C might reach ~50%. The hybrid approach can significantly surpass this by tapping the energy forms that thermal cycles simply cannot reach (high-grade kinetic energy).

For clarity, **Table 1** provides a qualitative comparison of different conversion configurations, based on literature values:

Configuration	Direct Conversion (η_DEC)	Thermal Cycle (η_th)	Net Efficiency (η_total)
Rankine-only (baseline)	0% (none)	40% (steam, 550 °C)	**~40%** – Conventional limit[5]
Advanced Brayton-only	0% (none)	50% (He/sCO_2, 1100 °C)	**~50%** – High-temp reactor[11]
DEC-only (ideal aneutronic)	~70% (multi-modal)	0%	**~70%** – Theoretical target[4]
Hybrid DEC + Brayton	50% (direct collectors + radiovoltaic)	50% (Brayton on rest)	**~75%** – Potential in D–^{3}He[51]

Table 2: Typical efficiencies for different conversion approaches in fusion reactors. The hybrid case shows how a 50% direct conversion combined with a 50% efficient Brayton on remaining heat can yield ~75% overall. In practice, the balance depends on fraction of energy in charged particles/photons and engineering limits of each subsystem. (Sources: direct conversion projections[4], advanced Brayton studies[51], Apollo/ARIES data[5][69]).

Integrating these systems requires careful engineering of the plant layout. The direct collectors (electrostatic grids or MHD channel) would likely be placed immediately at the fusion core's periphery or along divertor lines to capture energetic ions before they thermalize. These systems would need power electronics to combine and regulate their DC output. The thermionic converters might double as parts of the first wall or blanket, operating at high temperature and directly driving current out. The Brayton cycle would take coolant (helium or sCO_2) through heat exchangers that intercept any remaining heat – for example, the blanket that shields the core (which in aneutronic reactors can be much thinner since neutron flux is low) can be designed to run hot coolant to a turbine. Any **neutrons (for D–^{3}He, a few percent of power)** can be captured in a small thermal blanket that also couples to the coolant. The fact that aneutronic designs like Apollo-L3 could have a *permanent first wall* (no frequent replacement)[70] means one can integrate long-lived energy conversion surfaces (like thermionic emitters or PV arrays) into the reactor structure without excessive downtime.

One crucial integration aspect is **load management**: DEC systems respond almost instantaneously (on microsecond to millisecond scales for changes in plasma output) since they directly convert kinetic energy. A Brayton turbine has inertia and a slower response. Managing transients in a hybrid system will require power electronics and possibly energy storage (capacitors, flywheels) to buffer the direct output. On the other hand, the combination provides **flexibility**: during pulse operation or startup, one might not engage all direct converters, routing more energy to heat if needed to stabilize the plasma or maintain minimum turbine power. Advanced control could dynamically optimize the split η_DEC vs (1–η_DEC) to maximize efficiency at varying loads. The integration also allows **cogeneration**: for example, a Brayton loop might also be used for direct thrust in a spacecraft or for process heat in an industrial plant, while DEC provides electricity.

From a technology point of view, each subsystem in the hybrid will operate at extreme conditions: e.g., MHD channels in multi-tesla fields at thousands of K, thermionic emitters at 1600 K, Brayton turbines potentially needing ceramic blades. Development in those areas (high-B superconducting

magnets, advanced ceramics and high-temperature alloys, vacuum electronics) will feed directly into making hybrid fusion DEC feasible. The Department of Energy has ongoing programs examining sCO_2 Brayton loops for fusion pilot plants (aiming to demonstrate >45% efficiency power conversion)[71][72]. In parallel, private fusion companies interested in D–^{3}He (such as Helion Energy) implicitly rely on direct electrical output from their pulsed FRC approach[73] – essentially an inductive direct conversion. These efforts will synergize in creating a full hybrid system. Ultimately, a well-designed hybrid aneutronic reactor could achieve an **η_total in the range of 60–80%**, revolutionizing the economics of fusion by delivering far more electrical power per unit of fusion power generated than the ~30–40% of conventional neutron-driven designs.

Applications and Deployment Profiles

The unique advantages of direct energy conversion – high efficiency, high power density, and direct DC output – open up applications beyond the reach of traditional fusion or fission systems. We outline a few deployment scenarios:

- **Compact Generators for Space and Mobile Platforms:** Aneutronic DEC reactors are especially attractive for **spacecraft propulsion and power**. Without heavy turbine machinery and with greatly reduced radiation shielding (thanks to low neutron output), a D–^{3}He or p–^{11}B reactor with direct conversion could achieve very high specific power (W/kg). For instance, a fusion-powered spacecraft might use direct MHD or electrostatic converters to supply electricity to electric thrusters. NASA studies have imagined D–^{3}He reactors with direct conversion achieving Q ~5–10 and producing electricity for ion drives[74][75]. The direct DC output is ideal for ion engines or plasma thrusters, which require high-voltage DC. By eliminating the "thermal middleman," one also avoids large radiators – a crucial benefit in space. **Figure 1** illustrates a concept of a spacecraft using radiovoltaic cells to directly convert reactor radiation to power, driving a plasma thruster.

Figure 10: Artist's concept of a aneutronic fusion spacecraft. High-power radiovoltaic arrays directly convert the fusion radiation to electricity, powering plasma thrusters (DARPA "Rads-to-Watts" concept). Such designs leverage DEC to bypass heavy turbine hardware in space[54][55].

On Earth, **mobile power units** (e.g. for remote sites, military forward bases, large ships) could benefit from compact aneutronic reactors. A p–^{11}B reactor producing, say, 10 MW could be made relatively compact if no massive shielding or secondary loops are needed. Direct conversion to electricity means fewer moving parts and potentially **faster start-stop cycling**, akin to a battery. Such a reactor-generator could provide on-demand power in the field with minimal heat signature (since most energy goes to electricity, not waste heat).

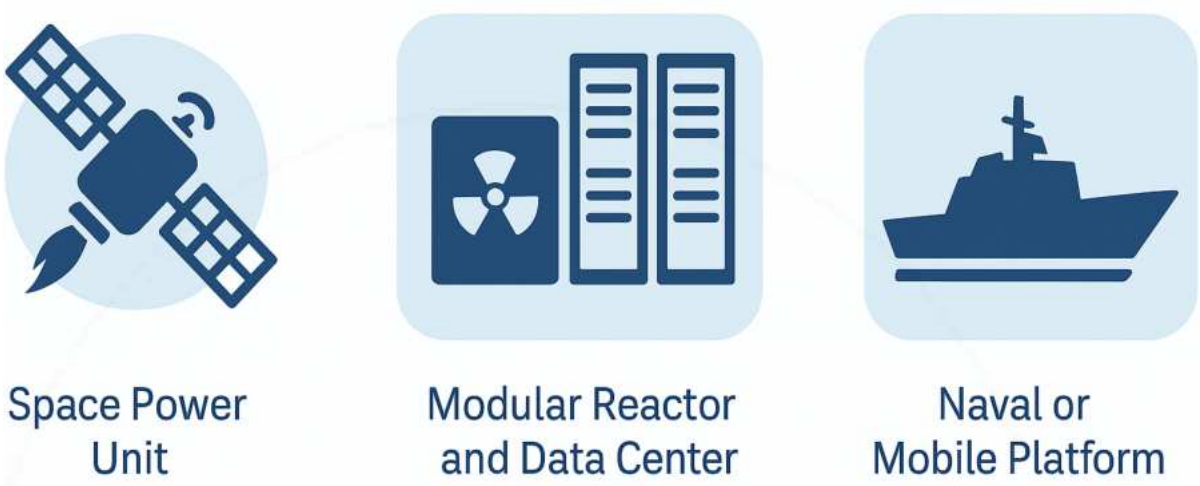

Figure 11: Representative deployment scenarios for the S.M.A.R.T. reactor: (left) compact space power unit, (center) modular reactor powering hyperscale data centers, and (right) naval or mobile power platform.

- **Grid-Scale Power (High-Value and Industrial Use):** The high efficiency of DEC reactors means less waste heat per MW – which eases cooling requirements and siting. A neutron-free (or neutron-poor) fusion plant can be located closer to population or industrial centers without radiation concerns. This opens the door to **grid-level deployment in urban areas or for dedicated industrial power**. One particularly relevant modern application is powering **AI and computational data centers**. These facilities have enormous and growing energy demands, and they value high-quality, reliable power. A aneutronic fusion plant with, say, 70% efficiency producing direct DC could be directly integrated with data center power infrastructure (which often already uses DC distribution internally). Moreover, the absence of large

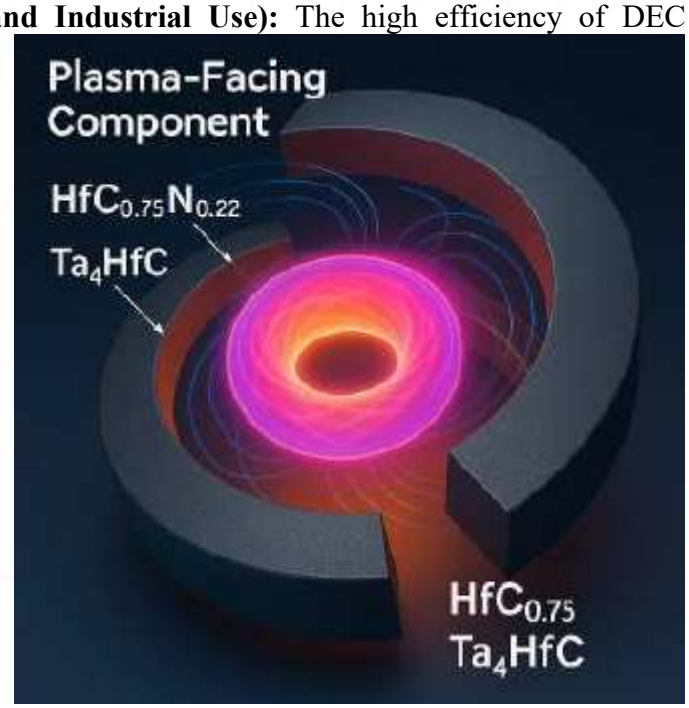

Figure 12: Hybrid energy conversion architecture combining direct electrostatic and MHD converters with a closed Brayton cycle bottoming loop. This approach raises net plant efficiency to >70 % while reducing thermal rejection load.

spinning turbines could allow more rapid **load-following** – direct converters can ramp output quickly by tuning plasma exhaust or bias voltages, which could match the fluctuating loads of computational clusters. Additionally, industries requiring high-temperature process heat (chemical plants, hydrogen production) could tap the Brayton waste heat of such a hybrid plant while the majority of energy goes into electricity for on-site use or export.

- **Multimodal Energy Shaping & Storage Integration:** A DEC-based fusion reactor inherently produces **multiple forms of energy** – electricity (from direct converters) and some heat (from Brayton or residual). This lends itself to **cogeneration and energy storage schemes**. For example, the direct electrical output could be split, with a portion going to the grid and a portion to charge battery banks or supercapacitors during times of surplus. The thermal output could feed an industrial process or a thermal storage medium (like molten salt) for later use. Because DEC can rapidly adjust (by controlling, say, the bias on electrostatic grids or the load on an MHD channel), the reactor can potentially modulate between more electrical output vs. letting more energy go to heat depending on demand. This is ideal for future grids with high renewable penetration, where a fusion plant might provide flexible backup – it could operate in a mode that charges a thermal storage when excess power exists, then ramp electrical output when solar/wind dip. Furthermore, aneutronic reactors can be more readily turned down or pulsed without the issues of maintaining a critical chain reaction (as in fission) or tritium breeding requirements.

Another intriguing application is using the direct fast proton output of D–^{3}He for **transmutation or medical isotope production**[76]. A beam of 14.7 MeV protons directly extracted could be directed to target materials (for medical isotope production) or for incinerating nuclear waste (though aneutronic reactors produce little waste themselves, they could help burn fission waste externally). This is a form of *direct use of fusion products*, essentially treating the reactor as a combined power plant and particle accelerator.

AI/ML load management: As hinted, the synergy between DEC fusion and AI-centric power usage is strong. Future AI training facilities might demand transient loads of many tens of MW. A DEC fusion plant's fast control and high power quality (DC output that can be converted to precise voltages) means it could be tightly managed by AI energy management systems. The plant could even be co-located with computing resources, minimizing transmission losses – something not feasible with today's gigawatt-scale, steam-cycle nuclear plants that must sit far from cities. The lack of long-lived radioactive waste in aneutronic fusion further simplifies deployment in populated or sensitive areas.

Finally, the **ultra-compact reactors** enabled by DEC (especially if p–^{11}B fuel became viable, as it requires no external neutron source or breeding) raise the prospect of **distributed fusion generators**. Instead of a few huge plants, one could imagine many 50–100 MWe units near consumers. This distributed model enhances grid resilience and reduces transmission needs. Each unit's high efficiency means less cooling infrastructure – possibly air-cooling could suffice for smaller units, using the fact that only ~20–30% of energy becomes waste heat (versus ~60–70% in a fission or D–T plant).

In summary, DEC in aneutronic reactors amplifies the already radical potential of fusion by delivering a form of energy output that is flexible, high-quality, and easier to integrate with modern energy systems. Whether for propelling spacecraft to Mars or powering an AI-driven economy on Earth, the ability to directly and efficiently convert fusion energy to electricity is a transformative

advantage. The coming together of advanced materials, power electronics, and fusion science in the next decade will determine how quickly these applications can become reality.

Risk Acknowledgment and Ongoing Research Disclosure

The innovative direct energy conversion (DEC) systems at the heart of Kronos's outer wall design present a set of formidable technical risks that the team is addressing with dedicated R&D and careful experimentation. Unlike conventional reactors that use well-understood thermal cycles, S.M.A.R.T.'s design relies on three novel DEC pathways – magnetohydrodynamic (plasma dynamo) converters, thermionic emitters, and photon-to-electric (photoelectric) converters – all integrated into the plasma-facing wall. We fully acknowledge that achieving the projected >60% direct electrical efficiency in practice[2][3] will require surmounting challenges in plasma–converter coupling, electrode/material survivability, and power electronics integration that have never been solved at reactor scale.

Materials and interface risks are paramount: the first-wall DEC modules (coated with LaB_6/CeB_6 thermionic films, graphene electrodes, advanced photovoltaics, etc.) must endure intense bombardment by 3.6 MeV alphas and hard UV/X-ray photons without degradation[5]. To tackle this, Kronos has initiated a materials testing program where candidate converter materials and coatings are exposed to plasma-like heat and particle flux in a specialized linear plasma device. Early results have guided improvements – for instance, a proprietary nano-layered graphene coating is now being applied to thermionic emitters to mitigate sputtering and maintain emissivity after thousands of pulses.

We are also leveraging insights from past research: experimental direct converters in the 1970s (e.g. the "Venetian blind" electrostatic collectors) demonstrated up to ~65% ion energy recovery in mirror machines[6], and the Apollo-L3 study in 1991 suggested that synchrotron radiation rectification could reach 80% efficiency[3]. Thermionic energy conversion, though demonstrated in laboratory settings, has not achieved the long-term stability required for reactor deployment[7]. These optimistic benchmarks from literature serve as proof-of-concept, but we treat them cautiously – the Kronos team is acutely aware that those results were at laboratory scale and often under idealized conditions.

Thus, each DEC subsystem in S.M.A.R.T. is being prototyped individually at increasing scales:

- **MHD converter**: we built a subscale plasma loop to verify that induced currents scale with our high-field predictions and to assess electrode erosion in flowing plasma.
- **Thermionic module**: we have a high-heat-flux electron gun test rig to evaluate emission currents from our LaB_6-based cathodes at 1000–1700 K, iterating designs for maximum output and longevity.
- **Photoelectric system**: we are testing multi-junction photovoltaic surfaces under simulated fusion photon spectra (using synchrotron facilities to emulate the extreme UV/X-ray range) to measure conversion efficiency and radiation tolerance.

These tests inform our models and de-risk the integrated system.

A major integration risk is the coupling between the plasma and DEC systems – for example, applying electrostatic fields or drawing currents could perturb the plasma itself. Kronos is mitigating this via careful engineering of field geometries (shaping the plasma edge with mirror

coils to direct exhaust flow into DEC channels) and by implementing AI-managed feedback loops that modulate the DEC operation to avoid plasma disturbances. Our real-time control system can, for instance, momentarily ramp down an MHD channel or bias voltage if sensors detect any adverse change in plasma confinement, then smoothly ramp back up once stable – ensuring that power extraction does not come at the expense of plasma stability[4]. Diagnostics on the wall (such as the graphene sensor network) will continuously monitor the health and output of DEC components, enabling predictive maintenance: any drop in a module's performance or any hot spot on the first wall triggers an automated adjustment or a scheduled replacement if needed.

Because these systems are first-of-a-kind, unknown unknowns remain a factor – to address this, we maintain a broad collaboration with academia and national labs, soliciting peer review of our DEC designs and even running scaled experiments (where feasible) at existing fusion devices to validate our concepts.

Kronos commits to a policy of transparency on this high-risk aspect of our design. By 2030, we will publish detailed results from our DEC prototype test program, including achieved efficiencies, failure modes encountered, and solutions implemented. We will report whether the integrated S.M.A.R.T. outer wall meets its design targets or if certain pathways fell short, along with the corrective actions or redundancy measures taken. The ongoing research disclosure will cover not just successes but also any setbacks in DEC development – for example, if a certain coating failed under neutron-equivalent radiation, we will document the finding and how we overcame it.

In summary, while the direct energy conversion approach is a game-changing feature of Kronos's reactor (enabling compact, high-efficiency power output), we approach it with scientific rigor and cautious optimism. Every layer of the outer wall is backed by a parallel R&D thread to ensure it can perform as intended. Proprietary risk-mitigation programs in materials science, high-voltage engineering, and plasma–surface interactions are running continuously to resolve these challenges. We affirm that no DEC module will be put into operation in the final device without proof from prototypes that it can operate reliably and safely under fusion-relevant conditions. By front-loading this work now and openly sharing our progress, we aim to retire the key risks of the outer wall and conversion system well ahead of our first commercial deployment.

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Conclusion

Aneutronic spherical tokamaks with direct energy conversion represent a bold leap toward **ultra-efficient fusion power**. By harnessing charged fusion products and high-energy photons *in situ*, these reactors sidestep the conventional thermal bottlenecks that have constrained power plants for over a century. The analysis in this paper highlights that through a combination of electrostatic collectors, MHD channels, thermionic and radiative converters – complemented by Brayton-cycle turbines – it is technically plausible to achieve net electric efficiencies on the order of 60–80% in advanced fusion systems[11][4]. This is more than double the efficiency of today's fission or projected D–T fusion plants, fundamentally altering the economic equation of fusion. Higher efficiency means less fuel consumption (e.g., costly ^{3}He or ^{11}B), smaller heat rejection systems, and higher power density – all critical for making fusion competitive and enabling new missions (like space travel).

The **strategic value** of DEC in aneutronic fusion goes beyond efficiency. It enables a reactor that is practically **neutron-free**, greatly reducing radiation damage and radioactive waste. A permanent first wall becomes feasible[70], and maintenance and downtime drop accordingly. This boosts the capacity factor and longevity of the plant. The **output is inherently in electrical form** (high-voltage DC), which is ideal for the increasingly electronics-driven world. High-voltage DC can be transmitted over long distances with low losses or converted to AC with power electronics, offering flexibility in grid integration. Additionally, direct DC is well-suited for certain high-end uses (datacenters, server farms, electrolysis hydrogen production, etc.) without extra conversion steps.

An aneutronic DEC reactor is also relatively **compact for its power output**. Eliminating the large steam turbine and associated cooling towers shrinks the plant footprint. This opens possibilities like powering isolated communities, providing ship propulsion (imagine a fusion-powered cargo ship or naval vessel with no refueling needed), or even mobile disaster relief power sources. The high efficiency means minimal cooling needs – potentially achievable with just air or modest water flow – making siting easier. Environmental impact is also minimal: no greenhouse gases, negligible radioactive waste, and far less thermal pollution due to the high conversion efficiency.

Perhaps one of the most futuristic but plausible outcomes of mastering this technology is **"AI-synchronous" power**: reactors whose output can dynamically respond to the demands of large-scale AI computations and industrial processes in real-time. Such agility is unheard of in current baseload plants. It will be enabled by the fine control afforded by direct energy extraction – tapping electrical power directly from the plasma allows fast control loops, potentially even using AI to modulate plasma and converter states in synchrony. The fusion plant of the future could thus become a smart energy node, allocating energy in real-time between computation, grid supply, and storage, under the guidance of advanced control algorithms.

In conclusion, direct energy conversion elevates aneutronic fusion from a scientific curiosity to a compelling solution for 21st-century energy needs. It addresses the historical shortcomings of fusion (inefficiency and large, complex plants) by **radically improving efficiency and power density**. Significant R&D challenges remain on the road to realization – from materials that can handle extreme conditions to integrated testing of DEC subsystems – but the path is increasingly illuminated by both historic successes (like the TMX 48% DEC test[27]) and modern innovations (e.g. radiovoltaics, high-T_n superconductors, nanomaterials). As we push into the late 2020s and 2030s, public and private fusion efforts are incorporating aneutronic principles, and DEC is

featuring more prominently in design studies. The Apollo-L3 study back in 1990 presciently showed the "compelling" merits of D–^{3}He reactors with direct conversion[77][78]. Now, with three more decades of technological progress, we are poised to turn that vision into reality. Achieving a **spherical tokamak aneutronic reactor with DEC** would herald a new era: compact, **ultra-clean fusion power plants** that directly empower our high-tech society, from Earth to deep space, with unparalleled efficiency and control[9][69]. The pursuit of this goal aligns scientific innovation with the pressing need for sustainable, high-performance energy – a synergy that could define energy generation for the century to come.

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TABLES

Table 1: Fusion energy conversion pathways, technologies, and efficiencies. Charged particle kinetic energy (α particles, protons) is captured via electrostatic direct collectors and MHD induction generators with ~60–80 % efficiency at TRL 3–4. Photon radiation (bremsstrahlung, synchrotron) can be converted using photoelectric PV (~10–20 %) or thermal Brayton cycles (~45–50 %), corresponding to TRL 2–3 for PV and 8–9 for Brayton. Residual thermal energy is recovered through mature Rankine steam turbine cycles (~30–35 % efficiency, TRL 9). 16
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FIGURES

Figure 1: High-level schematic of energy capture in the S.M.A.R.T. architecture. Charged particles are routed to electrostatic or MHD converters, photons are harvested via radiovoltaics and Brayton thermal loops, and residual heat is recovered through a Rankine cycle for maximal energy utilization (≥70 % potential net efficiency). 4
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Direct Energy Conversion in the S.M.A.R.T. Architecture

Performance, Modal Diversity, and System Integration

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